Nonlinear Dynamics in Cryptography

Logistic-map simulations and an educational interval-mapping text demonstration

Course Project

Abstract

This project is based primarily on Baptista's chaos-based encryption proposal [1], with Alvarez et al. [2] and Li et al. [3] used to assess its limitations. We study the logistic map through numerical simulations of its time evolution, bifurcations, Lyapunov exponent, sensitivity to initial conditions, and invariant density. We then test a small interval-mapping text demonstration. At r = 4, the numerical Lyapunov exponent is 0.693020, close to ln 2 = 0.693147; the mean of 500,000 retained samples is 0.4992599. Four test messages are recovered exactly. The text cipher is a simplified teaching model, not a faithful implementation of [1] and not a secure encryption method.

Primary research paper	Supporting literature
Baptista (1998): iteration-count encryption using the logistic map [1]	Alvarez et al. (2003): cryptanalysis [2]; Li et al. (2004): problems and countermeasures [3]

Keywords: logistic map; chaos; Lyapunov exponent; invariant density; interval mapping; cryptanalysis

1. Introduction and motivation

Encryption through chaos is interesting because a deterministic map can produce rapidly separating trajectories from nearby initial conditions. The key question is narrower than simply asking whether the orbit looks irregular: can an interval-visiting rule encode and recover text, and what limits prevent it from becoming a secure cipher? We use the logistic map because it is simple enough to analyse while still showing fixed points, period doubling, periodic windows, and chaotic regimes.

2. Model system and calculations

For the single-orbit studies we use r = 4.0 and x0 = 0.213. The bifurcation diagram scans 3,000 values of r in [2.8, 4.0], discards 1,000 transient iterations, and stores 200 states at each r. The Lyapunov sweep uses 3,000 r values in [2.5, 4.0], 1,000 transient iterations, and 3,000 accumulation iterations. The invariant-density calculation retains 500,000 states after a 1,000-step transient.

Quantity	Analytical result used in this project	Role in the simulation
Map	x_(n+1) = r x_n (1 - x_n)	Defines the orbit
Fixed points	x* = 0; x* = 1 - 1/r	Shows possible long-term states
Local stability	f'(x) = r(1 - 2x); stable when |f'(x*)| < 1	Explains changes with r
Largest Lyapunov exponent	lambda = mean[ln |r(1 - 2x_n)|]	Positive values indicate average separation

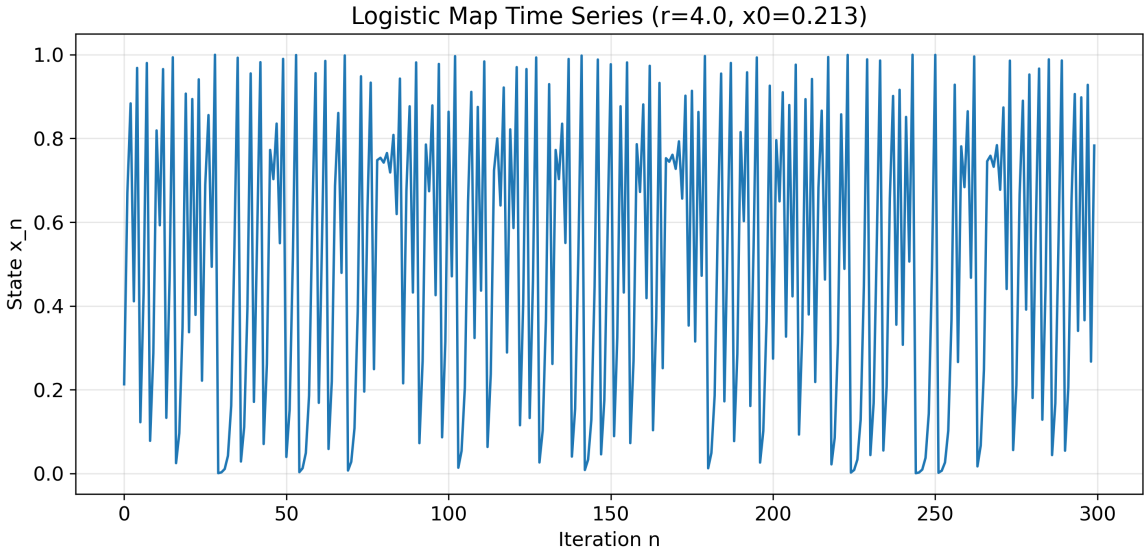


Figure 1. Numerical orbit at r = 4 and x0 = 0.213. The irregular appearance comes from a deterministic update rule.

3. Summary of results: bifurcations and sensitivity

The parameter sweep links the simple equation to its changing long-term behaviour. The Lyapunov calculation provides a quantitative check: positive values indicate average separation of nearby states. The two trajectory plots use starting values separated by only 10^-10.

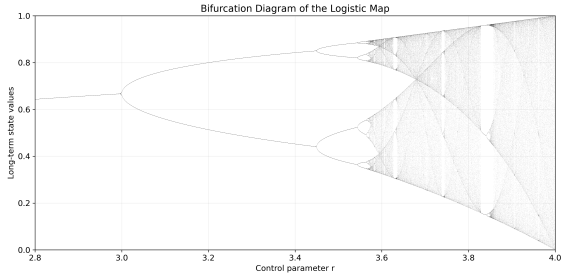


Figure 2. Bifurcation diagram after removing transients.

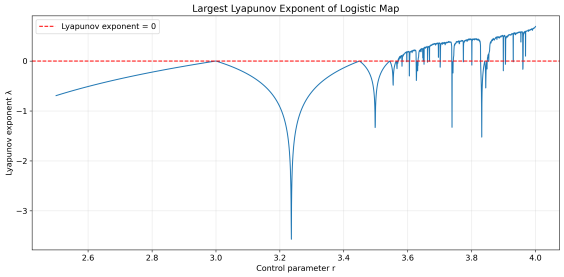


Figure 3. Largest Lyapunov exponent across the parameter sweep.

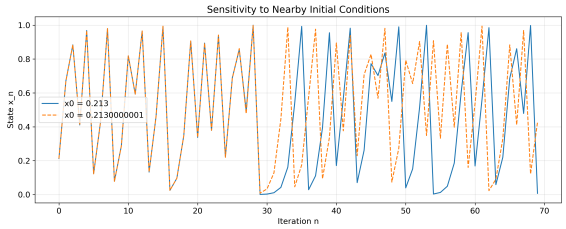


Figure 4. Nearby initial conditions separate under iteration.

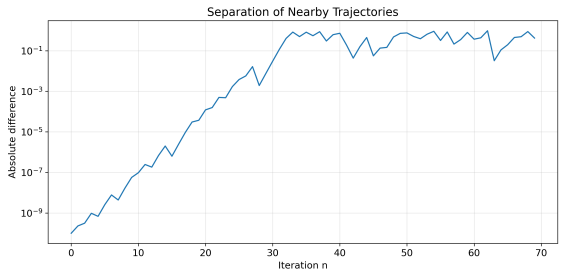


Figure 5. Absolute separation on a logarithmic scale.

Observed result	Numerical evidence	Interpretation
Chaotic parameter regime	At r = 4: lambda = 0.693020; ln 2 = 0.693147	Finite-time estimate agrees closely with the known value
Sensitive dependence	Initial separation 10^-10 becomes visible over 70 iterations	Illustrates instability, not randomness
Bifurcations	Multiple branches and windows in the r sweep	Long-term behaviour changes as r varies

Invariant density

At r = 4, the empirical state histogram has the expected U-shaped form. The retained-sample mean is 0.4992599, close to the theoretical mean 0.5. This is a numerical consistency check for the logistic-map simulation; it does not give a cryptographic security result.

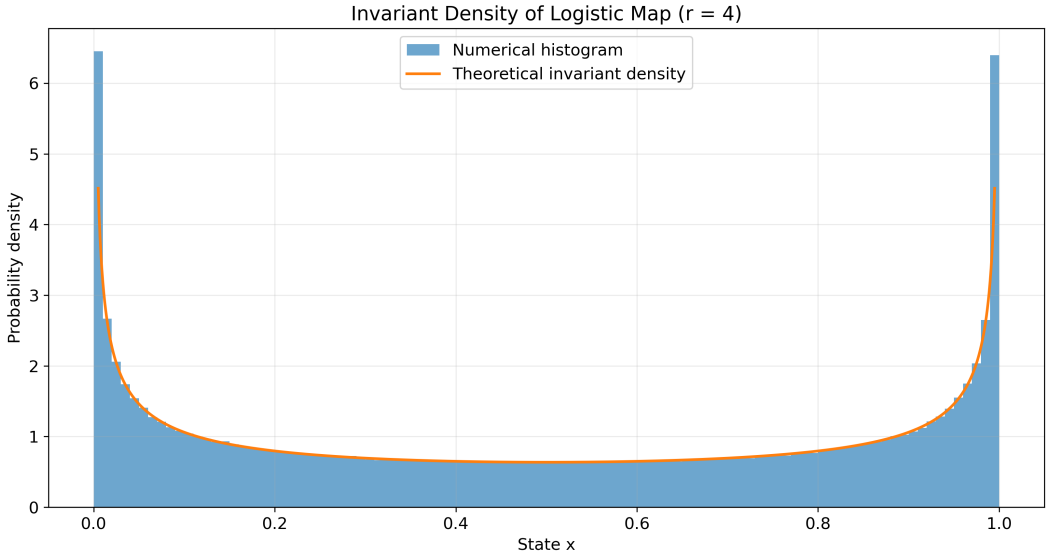


Figure 6. Numerical histogram and theoretical invariant density at r = 4.

4. Results: text recovery

The text experiment divides [0, 1] into 26 equal-width intervals for A-Z. For each character, encryption records the number of iterates required to reach the assigned interval. Decryption repeats that count from the same initial state. These tests verify consistency of this simplified implementation only.

Message	Characters	Recovered exactly	Measured time (s)
CHAOS	5	Yes	0.000212
LOGISTICMAP	11	Yes	0.000348
NONLINEARDYNAMICS	17	Yes	0.000620
ENCRYPTION	10	Yes	0.000360

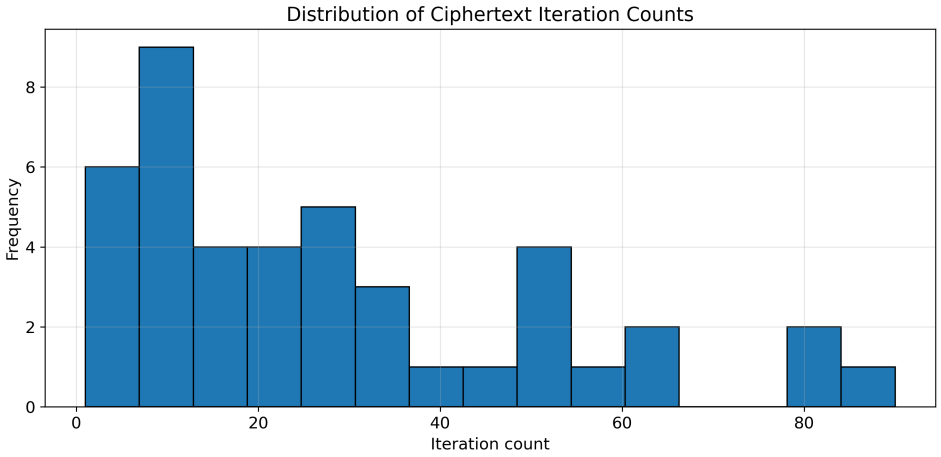


Figure 7. Distribution of the 43 iteration counts from the four text tests. This small histogram is descriptive, not a security test.

5. Encryption study, paper comparison, and limitations

Our demonstration follows the broad idea in [1]: character to interval, interval visit to iteration count, then count replay for decoding. The compact schematic below describes our code, not the complete published protocol.

Plaintext character	Assigned interval	Iteration count	Replay count and decode
e.g., A-Z	one of 26 equal intervals	stored as ciphertext	same r and initial state

What was not reproduced

Paper and result not reproduced	Why it was not reproduced in this project
Baptista [1]: complete original encryption protocol and its paper-specific worked results	The supplied code uses a fixed r = 4, one initial state, and 26 equal-width intervals over [0, 1]. It does not implement the original paper's full protocol choices or a reference test set.
Alvarez et al. [2]: cryptanalytic attack experiments on the ergodic cipher	No attack code, attack data, or controlled comparison with the original parameter choices was supplied. We therefore review these results rather than claim to reproduce them.
Li et al. [3]: numerical error behaviour and proposed countermeasures for Baptista-type systems	The project does not implement the modified schemes or a finite-precision error-propagation experiment. The code only tests reversible interval mapping in ordinary floating-point arithmetic.

6. Conclusion, outlook, and open questions

The simulations show the expected nonlinear behaviour of the logistic map, including sensitivity and a positive Lyapunov exponent at r = 4. The educational interval mapping recovers four short messages, but this proves only that the sender and receiver perform the same calculation. It does not establish confidentiality, attack resistance, or suitability for real data.

A useful next step is to implement the original protocol from [1] with its documented parameter choices, then reproduce one clearly specified result from [2] or [3]. Further work should measure ciphertext expansion, execution time, distributional bias, and finite-precision repetition, and compare the demonstration against a standard authenticated encryption scheme.

Code and reproducibility: https://github.com/Ashok-kumar-banoth/PH567_NLP_Project. The repository contains the submitted Colab export, the original preserved script, all seven figures, the README, and this report source.

Bibliography and references

[1] M. S. Baptista, "Cryptography with chaos," *Physics Letters A*, 240(1-2), 50-54 (1998). doi:10.1016/S0375-9601(98)00086-3

[2] G. Alvarez, F. Montoya, M. Romera, and G. Pastor, "Cryptanalysis of an ergodic chaotic cipher," *Physics Letters A*, 311(2-3), 172-179 (2003). doi:10.1016/S0375-9601(03)00469-9

[3] S. Li, G. Chen, K.-W. Wong, X. Mou, and Y. Cai, "Baptista-type chaotic cryptosystems: Problems and countermeasures," *Physics Letters A*, 332(5-6), 368-375 (2004). doi:10.1016/j.physleta.2004.09.028

Code availability

Project repository: https://github.com/Ashok-kumar-banoth/PH567_NLP_Project